WoSM Academy

MAT231-F27 · Calculus II

Sequences and series of functions

Pointwise and uniform convergence

Dr. Ali Khudiyev
WoSM Academy

Lecture 04 · Unit 4 · Fall 2027

CC BY-SA 4.0

Last time

- $\sum a_k$ converges **absolutely** if $\sum |a_k|$ converges; then it converges.
- **Alternating series test**: if b_k decreases to 0, then $\sum (-1)^{k+1} b_k$ converges, and $|S - S_n| \leq b_{n+1}$.
- **Conditionally** convergent series, such as $\sum \frac{(-1)^{k+1}}{k}$, can be rearranged to any sum. Absolutely convergent ones cannot.

So far every term a_k was a number. Today the terms are *functions* of x.

This lecture

01 From numbers to functions

02 Pointwise convergence

03 Uniform convergence

04 What uniform convergence preserves

05 Series of functions

06 Summary and practice

Reading: TODO(owner): choose a text for this unit — the books on the module's reading list do not cover uniform convergence.

01

From numbers to functions

Series whose terms depend on x

$$\sum_{k=0}^{\infty} x^k = \frac{1}{1-x}, \qquad e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}, \qquad \sum_{k=1}^{\infty} \frac{\sin(kx)}{k^2}$$

Each partial sum is a polynomial or a trigonometric sum: continuous, easy to differentiate and to integrate. The sum is a limit of such functions.

Three questions about a limit $f = \lim f_n$:

1. If every f_n is continuous, is f continuous?
2. Is $\int_a^b f = \lim \int_a^b f_n$?
3. Is $f' = \lim f_n'$?

For finite sums the answer is always yes. For limits it is not. Today: a condition that makes it yes. Units 5–7 depend on it.

02

Pointwise convergence

Pointwise convergence

Definition

Let $f_n\colon D \to \mathbb{R}$ for $n = 1, 2, \ldots$ The sequence (f_n) **converges pointwise** to $f\colon D \to \mathbb{R}$ if $f_n(x) \to f(x)$ for every $x \in D$: for every $x \in D$ and $\varepsilon > 0$ there is an N such that

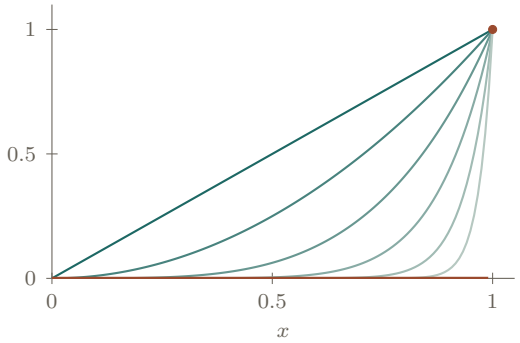
$$|f_n(x) - f(x)| < \varepsilon \qquad \text{for all } n \geq N.$$

Here N may depend on ε *and on* x.

Example

$\frac{x}{n} \to 0$ on $\mathbb{R}$. $\qquad \left(1 + \frac{x}{n}\right)^n \to e^x$ on $\mathbb{R}$ (Calculus I).

Continuity can be lost



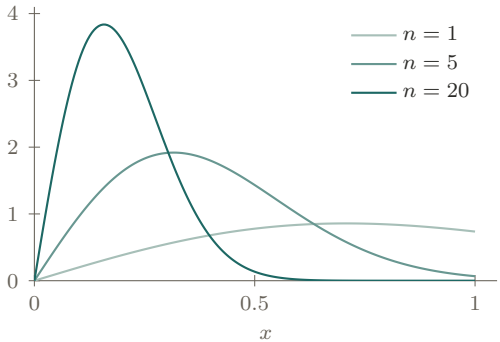
$f_n(x) = x^n$ on $[0,1]$:

$$f(x) = \lim_{n\to\infty} x^n = \begin{cases} 0, & 0 \le x < 1, \\ 1, & x = 1. \end{cases}$$

Every f_n is continuous; the limit f is not.

For $x^n < \varepsilon$ we need $n > \frac{\ln \varepsilon}{\ln x}$, which grows without bound as $x \to 1$: no single N works for all x.

Integrals can be lost



$f_n(x) = 2nx\,e^{-nx^2}$ on $[0, 1]$.

- $f_n(0) = 0$, and for $x > 0$, $f_n(x) \to 0$: the limit is $f = 0$.
- But $\int_0^1 f_n(x)\,\mathrm{d}x = 1 - e^{-n} \to 1$, as $-e^{-nx^2}$ is an antiderivative.

So $\lim \int_0^1 f_n = 1 \neq 0 = \int_0^1 \lim f_n$.
The bump gets narrower and taller; its area stays near 1.

Derivatives can be lost

Example

$f_n(x) = \dfrac{\sin(nx)}{n}$ on $\mathbb{R}$. Since $|f_n(x)| \le \frac{1}{n}$, $f_n \to 0$.

- $f_n'(x) = \cos(nx)$, so $f_n'(0) = 1$ for every n, but the limit 0 has derivative 0.
- At $x = \pi$: $f_n'(\pi) = (-1)^n$ does not converge at all.

The common cause

In each example N depends on x, and near some point it grows without bound. Two limits are being swapped — $n \to \infty$ and $x \to x_0$, or $n \to \infty$ and an integral — and pointwise convergence does not allow that.

03

Uniform convergence

Uniform convergence

Definition

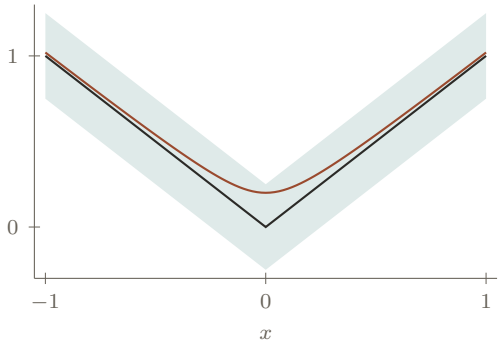
(f_n) **converges uniformly** to f on D if for every $\varepsilon > 0$ there is an N such that

$$|f_n(x) - f(x)| < \varepsilon \qquad \text{for all } n \geq N \text{ and all } x \in D.$$

Equivalently: $\sup_{x \in D} |f_n(x) - f(x)| \to 0$.

- Pointwise: for every x and every ε there is an N.
  Uniform: for every ε there is an N that works for *all* x.
- Uniform convergence implies pointwise convergence, to the same limit.
- It depends on the domain D: shrinking D can only help.

A picture



$f(x) = |x|$, the band $f \pm \varepsilon$ with $\varepsilon = \frac{1}{4}$, and $f_5(x) = \sqrt{x^2 + \frac{1}{25}}$.

Uniform convergence: from some N on, the *whole graph* of f_n lies inside the band, however narrow the band is.

Here $0 \le \sqrt{x^2 + \frac{1}{n^2}} - |x| \le \frac{1}{n}$ for all x, so $f_n \to |x|$ uniformly on $\mathbb{R}$.

Computing the supremum

- $\frac{x}{n}$: on $[0,1]$, $\sup = \frac{1}{n} \to 0$: uniform. On $\mathbb{R}$, $\sup = \infty$: not uniform.
- x^n: on $[0,a]$ with $a < 1$, $\sup = a^n \to 0$: uniform. On $[0,1)$, $\sup = 1$ for every n: not uniform.
- $2nx\,e^{-nx^2}$: the maximum is at $x = \frac{1}{\sqrt{2n}}$, with value $\sqrt{2n}\,e^{-1/2} \to \infty$: not uniform on $[0,1]$.
- $\frac{\sin(nx)}{n}$: $\sup = \frac{1}{n} \to 0$: uniform on $\mathbb{R}$.

Method

Find $\sup_x |f_n(x) - f(x)|$ (derivative $= 0$, or a bound); does it tend to 0? To show convergence is *not* uniform, points x_n with $|f_n(x_n) - f(x_n)| \not\to 0$ are enough.

The uniform Cauchy criterion

Theorem

(f_n) converges uniformly on D if and only if for every $\varepsilon > 0$ there is an N such that $|f_n(x) - f_m(x)| < \varepsilon$ for all $n, m \geq N$ and all $x \in D$.

Proof.

($\Leftarrow$) For each x, $(f_n(x))$ is a Cauchy sequence of numbers, so it has a limit; call it $f(x)$. Let $m \to \infty$ in $|f_n(x) - f_m(x)| < \varepsilon$: $|f_n(x) - f(x)| \leq \varepsilon$ for all $n \geq N$ and all x. ($\Rightarrow$) $|f_n - f_m| \leq |f_n - f| + |f - f_m|$. □

It proves uniform convergence without knowing the limit. The M-test (Section 5) uses it.

04

What uniform convergence preserves

Continuity

Theorem

If every f_n is continuous on D and $f_n \to f$ uniformly on D, then f is continuous on D.

Proof.

Let $x_0 \in D$ and $\varepsilon > 0$. Choose N with $|f_N(x) - f(x)| < \frac{\varepsilon}{3}$ for all $x \in D$, then $\delta > 0$ with $|f_N(x) - f_N(x_0)| < \frac{\varepsilon}{3}$ for $|x - x_0| < \delta$. For such x:

$$|f(x) - f(x_0)| \le |f(x) - f_N(x)| + |f_N(x) - f_N(x_0)| + |f_N(x_0) - f(x_0)| < \varepsilon. \quad \square$$

So x^n cannot converge uniformly on $[0, 1]$: its limit is not continuous.

Integrals

Theorem

If every f_n is continuous on $[a,b]$ and $f_n \to f$ uniformly on $[a,b]$, then $\int_a^b f_n(x)\,\mathrm{d}x \to \int_a^b f(x)\,\mathrm{d}x$.

Proof: f is continuous, hence integrable, and $\left|\int_a^b f_n - \int_a^b f\right| \le (b-a)\sup_{[a,b]}|f_n - f| \to 0$.

The interval must be bounded

$f_n(x) = \frac{1}{n}e^{-x/n} \to 0$ uniformly on $[0,\infty)$, as $\sup|f_n| = \frac{1}{n}$. But $\int_0^\infty f_n(x)\,\mathrm{d}x = 1$ for every n.

Derivatives

Theorem

Let every f_n be continuously differentiable on $[a,b]$. If $f_n' \to g$ uniformly on $[a,b]$ and $(f_n(x_0))$ converges for some $x_0 \in [a,b]$, then (f_n) converges uniformly to a function f, and $f' = g$.

Proof.

By the Fundamental Theorem of Calculus, $f_n(x) = f_n(x_0) + \int_{x_0}^{x} f_n'(t)\,\mathrm{d}t$. With $c = \lim f_n(x_0)$ and $f(x) = c + \int_{x_0}^{x} g(t)\,\mathrm{d}t$: $|f_n(x) - f(x)| \le |f_n(x_0) - c| + (b-a)\sup|f_n' - g| \to 0$ for all x. g is continuous (a uniform limit), so $f' = g$. □

The condition is on the *derivatives*: $\frac{\sin(nx)}{n} \to 0$ uniformly is not enough.

05

Series of functions

Series of functions

$\sum f_k$ converges pointwise or uniformly on D if its partial sums $S_n(x) = \sum_{k=1}^{n} f_k(x)$ do. Applied to S_n, Section 4 gives:

- **Continuity**: if every f_k is continuous and $\sum f_k$ converges uniformly, then $\sum f_k$ is continuous.
- **Integration term by term**: under the same conditions on $[a, b]$,

$$\int_a^b \sum_{k=1}^{\infty} f_k(x)\,\mathrm{d}x = \sum_{k=1}^{\infty} \int_a^b f_k(x)\,\mathrm{d}x.$$

- **Differentiation term by term**: if $\sum f_k'$ converges uniformly on $[a, b]$ and $\sum f_k(x_0)$ converges, then $\left(\sum f_k\right)' = \sum f_k'$.

The Weierstrass M-test

Theorem (Weierstrass M-test)

If $|f_k(x)| \le M_k$ for all $x \in D$ and all k, and $\sum M_k$ converges, then $\sum f_k$ converges absolutely and uniformly on D.

Proof.

For $n > m$ and every $x \in D$: $|S_n(x) - S_m(x)| \le \sum_{k=m+1}^{n} |f_k(x)| \le \sum_{k=m+1}^{n} M_k$, which is below ε for $m \ge N$ (Cauchy criterion for $\sum M_k$), for all x at once. Apply the uniform Cauchy criterion. □

It is the comparison test, made uniform: the bounds M_k do not depend on x.

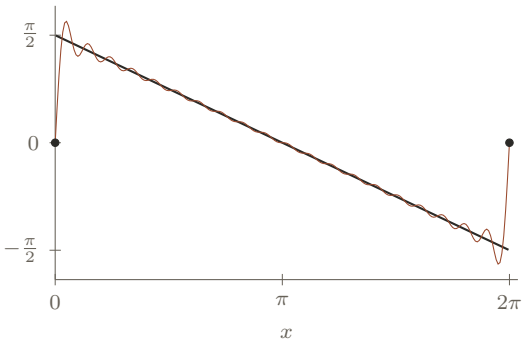
Example: $\sum \frac{\sin(kx)}{k^2}$

- $\left|\frac{\sin(kx)}{k^2}\right| \le \frac{1}{k^2} = M_k$ and $\sum \frac{1}{k^2}$ converges. So the series converges uniformly on $\mathbb{R}$, and its sum is a continuous function.
- We may integrate term by term:

$$\int_0^\pi \sum_{k=1}^{\infty} \frac{\sin(kx)}{k^2}\,\mathrm{d}x = \sum_{k=1}^{\infty} \frac{1-\cos(k\pi)}{k^3} = \sum_{j=1}^{\infty} \frac{2}{(2j-1)^3}.$$

- May we differentiate term by term? The derivative series is $\sum \frac{\cos(kx)}{k}$. At $x = 0$ it is the harmonic series, so it does not even converge everywhere. The theorem does not apply.

Pointwise but not uniform



$\sum_{k=1}^{\infty} \frac{\sin(kx)}{k}$ converges for every x, to $\frac{\pi - x}{2}$ for $0 < x < 2\pi$ and to 0 at $x = 0$ and $x = 2\pi$. (A Fourier series; we do not prove this here.)

Each partial sum is continuous, the sum jumps at 0: so the convergence is *not* uniform on $[0, 2\pi]$. In red: the partial sum with 20 terms, which overshoots near the jumps.

Geometric series as functions

For $|x| < 1$: $\displaystyle\sum_{k=0}^{\infty} x^k = \frac{1}{1-x}$.

- On $[-a, a]$ with $0 < a < 1$: $|x^k| \le a^k = M_k$ and $\sum a^k$ converges. So the convergence is uniform on $[-a, a]$.
- On $(-1, 1)$ it is not uniform: for each n the remainder $\sum_{k>n} x^k = \frac{x^{n+1}}{1-x}$ is unbounded as $x \to 1$.

> **Remark**
>
> That is enough. Every x in $(-1, 1)$ lies in some $[-a, a]$ with $a < 1$, and continuity, integrals over $[0, x]$ and derivatives at x only need uniform convergence on such an interval. The same holds for every power series inside its interval of convergence (Unit 5).

Integrating term by term

For $|t| \le |x| < 1$: $\dfrac{1}{1+t} = \sum_{k=0}^{\infty} (-t)^k$, uniformly between 0 and x (M-test with $M_k = |x|^k$). Integrate from 0 to x:

$$\ln(1+x) = \sum_{k=0}^{\infty} (-1)^k \frac{x^{k+1}}{k+1} = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots \qquad (|x| < 1).$$

- A power series for $\ln(1+x)$, from a geometric series and one theorem.
- At $x = 1$ the right-hand side is $1 - \frac{1}{2} + \frac{1}{3} - \cdots$, which converges (Lecture 3). That its sum is $\ln 2$ needs one more argument: Unit 6.

Differentiating term by term

On $[-a, a]$, $a < 1$: $|kx^{k-1}| \le ka^{k-1} = M_k$, and $\sum ka^{k-1}$ converges (ratio $\to a < 1$). So for $|x| < 1$:

$$\sum_{k=1}^{\infty} kx^{k-1} = \frac{\mathrm{d}}{\mathrm{d}x}\,\frac{1}{1-x} = \frac{1}{(1-x)^2}.$$

Example (Earlier series, now summed)

- Lecture 1, expected number of trials until the first success: $\sum_{k=1}^{\infty} k(1-p)^{k-1}p = \frac{p}{p^2} = \frac{1}{p}$.
- Lecture 2, where we only knew it converges: $\sum_{k=1}^{\infty} \frac{k}{2^k} = \frac{1}{2}\sum_{k=1}^{\infty} k\left(\frac{1}{2}\right)^{k-1} = \frac{1}{2}\cdot 4 = 2$.

Continuous, but nowhere differentiable

For $0 < a < 1$ and $b > 0$ let $W(x) = \sum_{k=0}^{\infty} a^k \cos(b^k \pi x)$.

- $|a^k \cos(b^k \pi x)| \le a^k$: by the M-test the series converges uniformly on $\mathbb{R}$, so W is continuous.
- Weierstrass showed in 1872 that if b is an odd integer and $ab > 1 + \frac{3\pi}{2}$, then W is differentiable at no point at all.
- The derivative series $-\sum a^k b^k \pi \sin(b^k \pi x)$ has coefficients $(ab)^k \pi \to \infty$ when $ab > 1$: no M-test applies, and term-by-term differentiation fails.

Uniform limits of smooth functions are continuous — and that is all they are guaranteed to be.

06

Summary and practice

Summary

- **Pointwise**: $f_n(x) \to f(x)$ for each x; N may depend on x. It can lose continuity, integrals and derivatives.
- **Uniform**: $\sup_{x \in D} |f_n(x) - f(x)| \to 0$; one N for all x.
- Uniform limits of continuous functions are continuous; on $[a, b]$ we may swap lim and $\int$.
- Derivatives: if $f_n' \to g$ uniformly and $f_n(x_0)$ converges, then $(\lim f_n)' = g$.
- **M-test**: $|f_k(x)| \le M_k$ with $\sum M_k < \infty$ gives uniform and absolute convergence of $\sum f_k$.
- The geometric series converges uniformly on every $[-a, a]$ with $a < 1$; we can integrate and differentiate it term by term inside $(-1, 1)$.

Practice

1. Find the pointwise limit, and decide whether the convergence is uniform:
   (a) $f_n(x) = \dfrac{x}{1+nx}$ on $[0,\infty)$
   (b) $f_n(x) = \dfrac{nx}{1+n^2x^2}$ on $[0,1]$
   (c) $f_n(x) = x^n(1-x)$ on $[0,1]$
   (d) $f_n(x) = e^{-nx}$ on $[0,\infty)$, and on $[1,\infty)$
2. Compute $\lim_{n\to\infty} \int_0^1 n^2 x\, e^{-nx}\,\mathrm{d}x$ and $\int_0^1 \lim_{n\to\infty} n^2 x\, e^{-nx}\,\mathrm{d}x$. What does this tell you?

Practice, continued

3. Show that $f(x) = \sum_{k=1}^{\infty} \frac{\cos(kx)}{k^3}$ is differentiable on $\mathbb{R}$, and find f'.
4. Show that $\sum_{k=1}^{\infty} \frac{x^k}{k} = -\ln(1-x)$ for $|x| < 1$. What does this give at $x = \frac{1}{2}$?
5. Find $\sum_{k=1}^{\infty} \frac{k^2}{2^k}$. Hint: differentiate $\sum_{k=1}^{\infty} kx^k = \frac{x}{(1-x)^2}$.
6. True or false: if every f_n is continuous, $f_n \to f$ pointwise, and f is continuous, then the convergence is uniform.

Answers

1. (a) 0; uniform: $0 \le \frac{x}{1+nx} < \frac{1}{n}$. (b) 0; not uniform: $f_n(\frac{1}{n}) = \frac{1}{2}$. (c) 0; uniform: the maximum, at $x = \frac{n}{n+1}$, is at most $\frac{1}{n+1}$. (d) 1 at $x = 0$ and 0 for $x > 0$: not uniform on $[0, \infty)$ (the limit is not continuous); uniform on $[1, \infty)$, where $\sup = e^{-n}$.
2. With $u = nx$ the integral is $\int_0^n ue^{-u}\,\mathrm{d}u = 1 - (n+1)e^{-n} \to 1$, while the pointwise limit is 0. So the convergence is not uniform on $[0, 1]$ (indeed $f_n(\frac{1}{n}) = \frac{n}{e}$).
3. The derivative series $-\sum \frac{\sin(kx)}{k^2}$ converges uniformly on $\mathbb{R}$ ($M_k = \frac{1}{k^2}$), and the series converges at $x = 0$. So $f'(x) = -\sum_{k=1}^{\infty} \frac{\sin(kx)}{k^2}$.
4. Integrate $\sum_{k \ge 0} t^k = \frac{1}{1-t}$ from 0 to x. At $x = \frac{1}{2}$: $\sum \frac{1}{k\,2^k} = \ln 2$.
5. $\sum k^2 x^{k-1} = \frac{1+x}{(1-x)^3}$, so $\sum \frac{k^2}{2^k} = \frac{1}{2} \cdot 12 = 6$.
6. False: $2nx\,e^{-nx^2} \to 0$ on $[0, 1]$, but not uniformly.

Next time

Lecture 05 — Power series (Unit 5)

- Series $\sum c_k (x-a)^k$ and their radius of convergence
- Uniform convergence inside the interval of convergence
- Differentiating and integrating power series term by term

Reading: Strang & Herman, *Calculus, Volume 2*, Sections 6.1 and 6.2.

Questions: the module channel on Discord, or office hours, Monday–Friday, 10:00–17:00 (Baku time, UTC+4), ali.khudiyev@wosm.academy.